

A Survey on Regime-Aware Portfolio Analytics and ML-Based Strategy Evaluation for Indian Equity Markets

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ABSTRACT – The increasing complexity and volatility of financial markets have created a growing demand for intelligent systems capable of accurate stock prediction, adaptive trading, and efficient risk management. Traditional statistical methods and many existing financial platforms primarily focus on visualization and isolated backtesting, often lacking integrated regime-aware and risk-aware analytical capabilities. This survey reviews recent advancements in machine learning, deep learning, and reinforcement learning techniques applied to stock market prediction, algorithmic trading, market regime detection, portfolio optimization, and financial stress testing. The study examines widely used models such as Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), Autoencoders, and Deep Reinforcement Learning (DRL). Additionally, emerging approaches involving sentiment analysis, cross-market data integration, and regime clustering are discussed. Key challenges including backtest overfitting, data non-stationarity, and model generalization are also analyzed. The reviewed literature indicates that hybrid machine learning approaches can significantly improve forecasting accuracy and trading performance compared to traditional techniques, while challenges related to robustness and real-time adaptability remain important research areas.

KEY WORDS: Stock Market Prediction, Algorithmic Trading, Machine Learning, Deep Learning, Reinforcement Learning, Market Regime Detection, Financial Risk Management, Sentiment Analysis, Portfolio Optimization

1. INTRODUCTION

Financial markets are highly dynamic, nonlinear, and influenced by numerous economic, political, and psychological factors. Predicting stock price movements and evaluating investment strategies have therefore become major challenges in quantitative finance and financial analytics. Traditional statistical and econometric methods often struggle to model the complex temporal dependencies, volatility fluctuations, and regime shifts present in financial time-series data. As a result, machine learning and deep learning approaches have gained significant attention due to their ability to identify

hidden patterns and learn from large volumes of historical market data. Recent studies have demonstrated the effectiveness of models such as Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), Deep Neural Networks (DNN), and Reinforcement Learning (RL) in improving stock market forecasting and automated trading performance.

The rapid growth of algorithmic trading and AI-driven financial systems has also increased the importance of portfolio intelligence, market regime detection, and risk-aware strategy evaluation. Existing financial platforms primarily focus on chart visualization and isolated backtesting functionalities, often lacking integrated systems capable of combining market prediction, regime analysis, strategy recommendation, and downside risk forecasting within a unified framework. Furthermore, financial datasets are characterized by non-stationarity, autocorrelation, noise, and sudden regime transitions, which make reliable model evaluation particularly difficult.

Several recent studies have explored advanced approaches for addressing these challenges. Hybrid deep learning architectures combining LSTM, DNN, and sentiment analysis have shown improved predictive capability for stock market forecasting. Reinforcement learning techniques have also been applied to algorithmic trading for adaptive decision-making under uncertain market conditions. In parallel, unsupervised learning methods such as Hidden Markov Models (HMM), clustering techniques, and Wasserstein-distance-based regime detection have emerged as effective tools for identifying changing market conditions and improving trading strategy robustness.

Another major concern in financial machine learning is backtest overfitting, where models perform well on historical data but fail to generalize in real market environments. Recent research emphasizes the need for specialized validation frameworks, robust cross-validation techniques, and probabilistic risk analysis for reliable financial model evaluation.

This survey reviews recent advancements in machine learning-driven financial market systems with a focus on stock market prediction, algorithmic trading, market regime detection, portfolio intelligence, and risk forecasting. The study analyzes major methodologies, compares their strengths and limitations, and identifies current research gaps in

developing adaptive and robust financial analytics platforms.

2. METHODS

The methodological framework adopted in this survey follows a systematic literature review approach aimed at gathering, analyzing, and synthesizing existing research on stock market prediction, portfolio analytics, market regime detection, and strategy evaluation systems. The primary research question guiding this survey is: What methods and platforms currently exist for evaluating financial trading strategies under varying market conditions, and what gaps remain, particularly for Indian equity markets? The survey further investigates the limitations of existing financial platforms in providing integrated regime-aware and risk-aware portfolio intelligence.

2.1 Search Strategy - Literature search was conducted across IEEE Xplore, Google Scholar, arXiv, ScienceDirect, and ACM Digital Library using keyword combinations including “stock market prediction using LSTM,” “algorithmic trading machine learning,” “market regime detection,” “portfolio backtesting framework,” “risk-adjusted strategy evaluation,” and “Indian equity market analytics.” Commercial platforms such as Zerodha, Streak, Groww, TradingView, Smallcase, INDmoney, and QuantConnect were additionally evaluated through official documentation and comparative feature analysis. Research published between 2018 and 2025 was prioritized, along with selected foundational studies relevant to financial machine learning and regime analysis.

2.2 Inclusion and exclusion criteria - The survey included papers related to stock market prediction, algorithmic trading, market regime detection, portfolio evaluation, financial stress testing, and machine learning-based risk analysis. Studies involving LSTM, CNN, reinforcement learning, clustering algorithms, Hidden Markov Models, and hybrid deep learning architectures were prioritized due to their relevance to intelligent financial systems. Papers focused solely on theoretical finance without computational implementation, cryptocurrency-only systems lacking broader equity applicability, duplicate studies, and publications without quantitative evaluation metrics were excluded. Platform analysis was limited to analytical and backtesting capabilities rather than brokerage-specific operations.

2.3 Analysis Framework - The selected studies were evaluated based on methodology, datasets used, prediction accuracy, validation techniques, risk metrics, computational complexity, and adaptability to changing market conditions. Additional emphasis was placed on integration between prediction models, regime detection systems, portfolio backtesting, and strategy recommendation

frameworks. Existing financial platforms were comparatively analyzed based on support for strategy backtesting, portfolio-level evaluation, regime detection, risk-adjusted performance metrics, machine learning integration, Indian market compatibility, and accessibility for non-technical users. The final synthesis enabled identification of current technological gaps and established the foundation for a proposed integrated regime-aware portfolio intelligence framework.

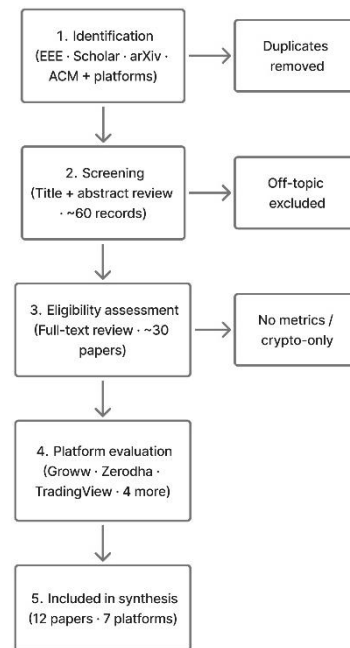


Fig-1: Survey Methodology Flowchart

3. RESULTS

The literature review identified five major research domains within intelligent financial market systems: stock market prediction, algorithmic trading, market regime detection, portfolio backtesting and risk evaluation, and existing financial analytics platforms.

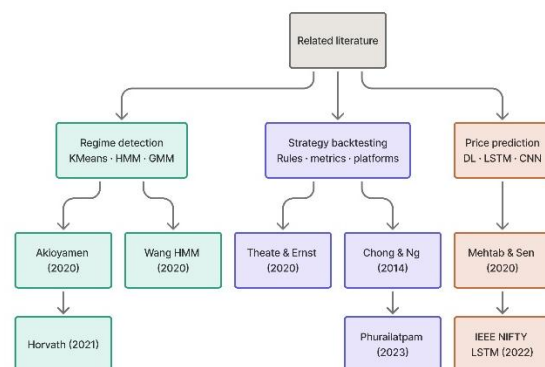


Fig-2: Classification of surveyed literature in intelligent financial systems.

The reviewed studies demonstrate that modern machine learning approaches significantly improve predictive capability and adaptive decision-making compared to traditional statistical methods. However, major limitations still exist in terms of integration, robustness, interpretability, and applicability to Indian equity markets.

3.1 Stock Market Prediction Models - A large portion of recent financial research focuses on predicting stock price movements using machine learning and deep learning techniques. Mehtab and Sen (2020) proposed an LSTM-DNN hybrid architecture for stock market prediction across multiple real-world datasets, demonstrating improved prediction accuracy compared to traditional forecasting methods. Similarly, Phurailatpam et al. (2022) utilized CNN-LSTM-based architectures for stock price prediction using historical financial indicators and time-series data.

Several studies also incorporated multi-market and cross-domain information into predictive frameworks. Cabrera et al. (2024) proposed a multi-market data-driven prediction framework that combined optimization techniques with deep learning architectures for enhanced forecasting performance across multiple stock exchanges. In another study, Wang et al. (2024) integrated social media sentiment analysis with Transductive Long Short-Term Memory (TLSTM) networks, demonstrating improved prediction capability under volatile market conditions.

Traditional technical indicators continue to remain relevant in financial forecasting. Chong and Ng (2008) revisited the performance of MACD and RSI oscillators, showing that momentum-based indicators still retain predictive significance under certain market conditions despite the rise of deep learning models.

Although these approaches achieved improved predictive accuracy, most systems remained limited to isolated forecasting tasks without integration into portfolio-level strategy evaluation or regime-aware investment analysis.

3.2 Algorithmic Trading and Reinforcement Learning - Recent advancements in reinforcement learning have enabled adaptive trading systems capable of learning optimal trading strategies directly from market environments. Moody and Saffell (2001) pioneered reinforcement learning applications in trading through direct reinforcement frameworks, while more recent work by Théate and Ernst (2020) applied Deep Reinforcement Learning (DRL) to algorithmic trading using Deep Q-Networks and policy optimization approaches.

Algorithmic trading systems commonly utilize OHLCV data, technical indicators such as RSI and MACD, and portfolio state variables for decision-making. Several studies demonstrated that reinforcement learning agents can outperform static rule-based strategies under simulated market

environments by adapting to changing price dynamics.

However, the reviewed literature also highlighted major practical limitations. Many reinforcement learning systems rely heavily on simplified reward structures, simulated trading environments, and computationally expensive training pipelines. Real-world deployment challenges such as transaction costs, slippage, overfitting, and changing market regimes remain insufficiently addressed.

3.3 Market Regime Detection Techniques - Market regime detection has emerged as an important research area for improving investment strategy evaluation and portfolio adaptability. Hidden Markov Models (HMM), K-Means clustering, Gaussian Mixture Models (GMM), and Wasserstein-distance-based clustering techniques have been widely explored for identifying bull, bear, and sideways market regimes.

Akiyamen et al. (2020) proposed unsupervised clustering approaches for identifying financial market states based on volatility and return characteristics. Wang et al. (2020) further demonstrated the effectiveness of Hidden Markov Models in regime-switching financial systems for adaptive portfolio allocation. Horvath et al. (2021) utilized Wasserstein distance-based clustering for improved separation of financial market regimes under non-stationary conditions.

Additionally, regime-aware factor investing frameworks using Hidden Markov Models were explored by several studies for tactical portfolio management and adaptive asset allocation. Hybrid learning approaches combining supervised and unsupervised methods also demonstrated improved adaptability for detecting financial regime switches.

Despite promising results, most existing regime detection systems operate independently from portfolio analytics and trading strategy recommendation frameworks, limiting their practical applicability for retail investors and non-technical users.

3.4 Portfolio Backtesting and Risk Evaluation - Robust portfolio backtesting and risk-adjusted evaluation remain critical components of financial machine learning systems. Several studies emphasized the importance of performance metrics such as Sharpe Ratio, Sortino Ratio, CAGR, volatility, and maximum drawdown for evaluating trading strategies under varying market conditions.

Lopez de Prado (2018) highlighted the growing problem of backtest overfitting in machine learning-based financial systems and introduced advanced validation techniques such as Purged K-Fold Cross-Validation and Combinatorial Purged Cross-Validation to improve out-of-sample reliability.

In parallel, stress-testing frameworks utilizing machine learning approaches were proposed for evaluating Indian financial market portfolios under

adverse economic conditions. Regime-aware portfolio optimization methods also demonstrated improved robustness by dynamically adapting trading strategies according to changing volatility and macroeconomic environments.

However, many existing systems provide only isolated backtesting functionality without integrating regime detection, probabilistic risk forecasting, or machine learning-driven strategy suitability analysis.

3.5 Comparative Analysis and Identified Gaps - Table 1 summarizes the major approaches identified across the surveyed literature, including stock market prediction, algorithmic trading, market regime detection, and portfolio evaluation systems. The reviewed studies demonstrate that deep learning architectures such as LSTM, CNN-LSTM, and hybrid DNN frameworks achieve improved forecasting accuracy compared to traditional statistical approaches. Reinforcement learning systems further improve adaptability by dynamically adjusting trading strategies under changing market conditions. Similarly, Hidden Markov Models, clustering algorithms, and Wasserstein-distance-based approaches have shown effectiveness in identifying market regimes and improving tactical portfolio allocation.

Despite these advancements, the literature reveals several important limitations. Most stock prediction systems focus only on forecasting accuracy without integrating strategy evaluation or portfolio-level analytics. Reinforcement learning frameworks often rely on simulated environments and computationally expensive training procedures, reducing their practical applicability in real-world trading. Regime detection models generally operate independently from backtesting systems and rarely provide interpretable investment insights for non-technical users. Additionally, existing financial platforms such as TradingView, Zerodha/Streak, and QuantConnect provide isolated functionalities such as chart visualization, manual backtesting, or algorithmic execution, but lack integrated regime-aware portfolio intelligence specifically designed for Indian equity markets.

Another major concern identified in the reviewed literature is the problem of backtest overfitting and poor out-of-sample generalization. Lopez de Prado (2018) demonstrated that many machine learning-based financial systems fail under non-stationary market conditions due to improper validation techniques and data leakage. Similarly, stress-testing frameworks for Indian financial portfolios remain limited in accessibility and integration with retail trading platforms.

The biggest gap identified across all surveyed domains is the absence of a unified framework capable of combining stock prediction, market regime detection, portfolio backtesting, machine learning-based strategy evaluation, and risk-

aware analytics within a single accessible platform. Most existing systems specialize in only one domain and do not provide integrated decision support for adaptive portfolio management under varying market conditions. This gap becomes even more significant in the context of Indian equity markets, where regime-aware retail analytics platforms remain limited.

Table 1: Comparative Summary of Surveyed Approaches

Approach	Method	Key Strength	Limitation
Stock Prediction	LSTM, CNN, DNN	Accurate forecasting	Limited integration
Algorithmic Trading	RL, Indicator Strategies	Adaptive trading	Overfitting risk
Regime Detection	HMM, K-Means, GMM	Detects market states	Standalone usage
Backtesting & Risk	Sharpe, Drawdown, CV	Strategy evaluation	Validation issues
Existing Platforms	TradingView, Zerodha	Visualization support	No regime analytics
Proposed Framework	ML + Regime + Backtest	Unified analysis	Needs real-world testing

3.6 Proposed Integrated System Overview - Based on the identified gaps, an integrated regime-aware portfolio intelligence framework is proposed that combines historical market data retrieval, market regime detection, portfolio backtesting, strategy evaluation, and risk analysis within a unified system. The framework utilizes Yahoo Finance data pipelines, technical indicators, and predefined strategies such as Buy-and-Hold, RSI-based trading, and Moving Average Crossover strategies. K-Means clustering is used to identify bull, bear, and sideways market conditions. The architecture combines a FastAPI backend with a Next.js frontend to provide interactive portfolio analytics, equity curve visualization, regime analysis, and ML-based strategy recommendations for Indian equity markets.

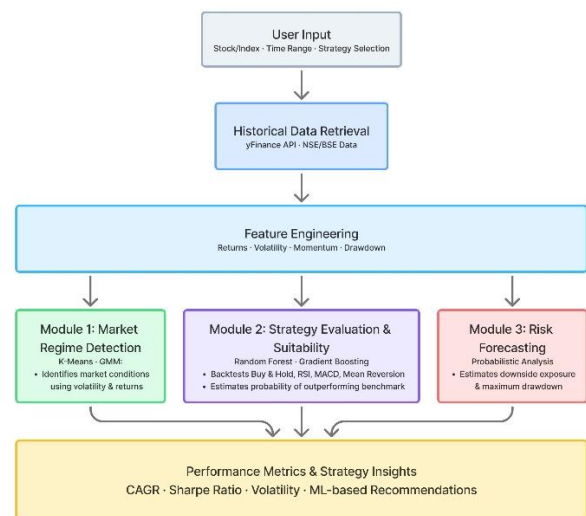


Fig. 2: Proposed Integrated System Architecture

Table 2: Expected System Performance Against Baseline

Metric	Baseline (Existing Platforms)	With Proposed System
Strategy Evaluation	Manual / isolated testing	Regime-aware evaluation
Market Analysis	Static indicators	Adaptive ML-based insights
Portfolio Backtesting	Limited strategy scope	Multi-strategy backtesting
Risk Assessment	Basic metrics only	Integrated risk analytics
Regime Detection	Not supported	Bull/Bear/Sideways detection
User Accessibility	Technical or fragmented	Unified interactive dashboard
Indian Market Support	Partial support	Indian equity focused
Strategy Recommendation	User-driven decisions	ML-assisted recommendations

4. DISCUSSION

The reviewed literature demonstrates that machine learning and deep learning techniques have significantly improved financial market prediction, trading strategy evaluation, and market regime analysis. However, research in this domain remains highly fragmented, with most studies focusing on isolated components such as stock forecasting, reinforcement learning-based trading, regime detection, or portfolio backtesting independently. Although each area has produced technically effective solutions, very few systems integrate these capabilities into a unified framework suitable for practical investment decision-making, particularly within Indian equity markets.

One major observation from the survey is that prediction accuracy alone is insufficient for building effective financial analytics systems. Several studies report strong forecasting performance using LSTM, CNN-LSTM, and reinforcement learning models, yet many of these approaches lack interpretability, regime awareness, and portfolio-level applicability under real market conditions. Additionally, financial markets are inherently non-stationary, making models highly sensitive to changing volatility patterns and economic conditions. As highlighted by Lopez de Prado (2018), improper validation methodologies can lead to severe backtest overfitting and unreliable out-of-sample performance.

Another important issue identified in the literature is the limited availability of integrated financial datasets. Real-world deployment of intelligent portfolio systems requires synchronized access to historical price data, macroeconomic

indicators, technical signals, volatility measures, and potentially sentiment-based information sources. However, many existing studies rely on simulated datasets, restricted market samples, or isolated benchmark environments, reducing practical generalizability. Similarly, commercial platforms provide strong visualization and charting support but rarely incorporate regime-aware machine learning pipelines or adaptive strategy recommendation systems.

The proposed framework addresses existing limitations by integrating market regime detection, portfolio backtesting, strategy evaluation, and risk-aware analytics within a unified architecture. K-Means clustering is used for regime identification due to its computational simplicity and interpretability, although future work may explore advanced probabilistic approaches such as Hidden Markov Models and Gaussian Mixture Models for improved regime consistency and real-time adaptation. Unlike existing platforms focused primarily on isolated visualization or manual backtesting, the proposed system combines predictive analytics with portfolio-level strategy evaluation and regime-aware decision support within a single accessible platform tailored for Indian equity markets.

5. CONCLUSION

The literature survey examined major research areas within intelligent financial market systems, including stock market prediction, algorithmic trading, market regime detection, portfolio backtesting, and risk-aware portfolio evaluation. The reviewed studies demonstrate that machine learning and deep learning approaches such as LSTM, reinforcement learning, clustering algorithms, and Hidden Markov Models significantly improve forecasting accuracy and adaptive trading performance compared to traditional statistical methods. However, most existing studies and commercial platforms remain focused on isolated functionalities such as prediction, visualization, or strategy testing without providing integrated regime-aware portfolio intelligence.

The survey identified major challenges including backtest overfitting, poor out-of-sample generalization, fragmented analytics pipelines, and limited accessibility of advanced quantitative tools for retail investors. Existing platforms such as TradingView, Zerodha/Streak, and QuantConnect provide strong charting and backtesting capabilities but lack integrated machine learning-driven systems tailored for Indian equity markets.

Based on the identified gaps, this paper proposed a regime-aware portfolio intelligence framework integrating market regime detection, portfolio backtesting, machine learning-based strategy evaluation, and risk analytics into a single accessible architecture. The proposed framework

demonstrates the feasibility of combining predictive analytics with adaptive strategy evaluation for Indian financial markets.

Future work may focus on incorporating real-time streaming market data, probabilistic regime-switching models, sentiment-aware forecasting, and advanced reinforcement learning strategies. Large-scale deployment and real-world validation across broader market conditions will also be necessary to evaluate scalability, robustness, and practical effectiveness. Overall, this survey establishes a structured understanding of current intelligent financial analytics systems and highlights the need for integrated regime-aware decision-support platforms for modern financial markets.

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